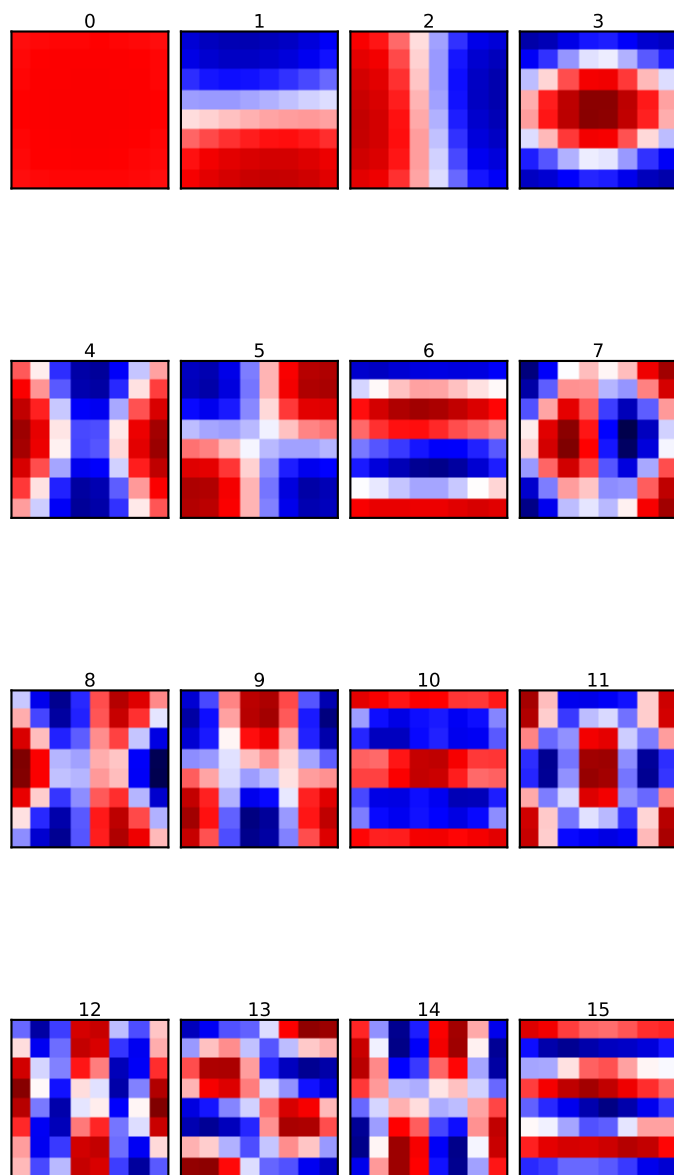
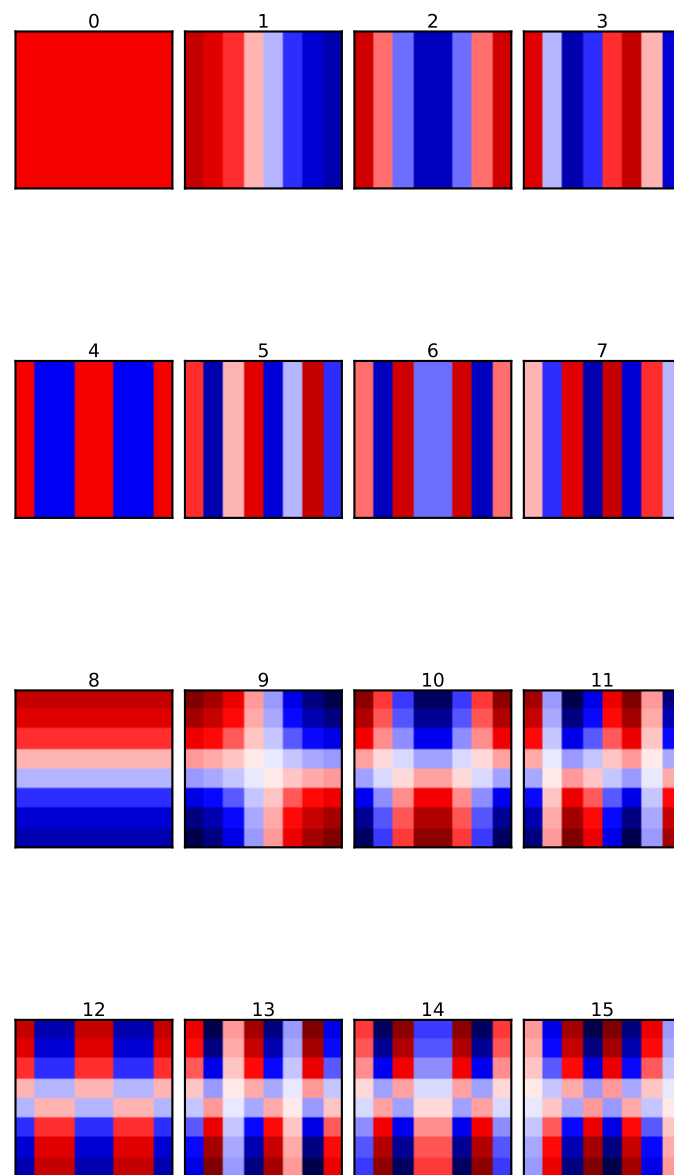


block\_PCA — top-16 eigen-patches (8×8)  
 rows of  $\Phi_b \in \mathbb{R}^{64 \times 64}$ , fit on 512000 patches



block\_DCT (closed-form) — first 16 basis patches (8×8)  
 $\Phi[k, \cdot] = \cos(\pi(n + \frac{1}{2})k/8)$  — for reference



global PCA — top-16 eigen-images (256×256)  
 rows of  $\Phi \in \mathbb{R}^{65536 \times 65536}$ , fit on 500 images

